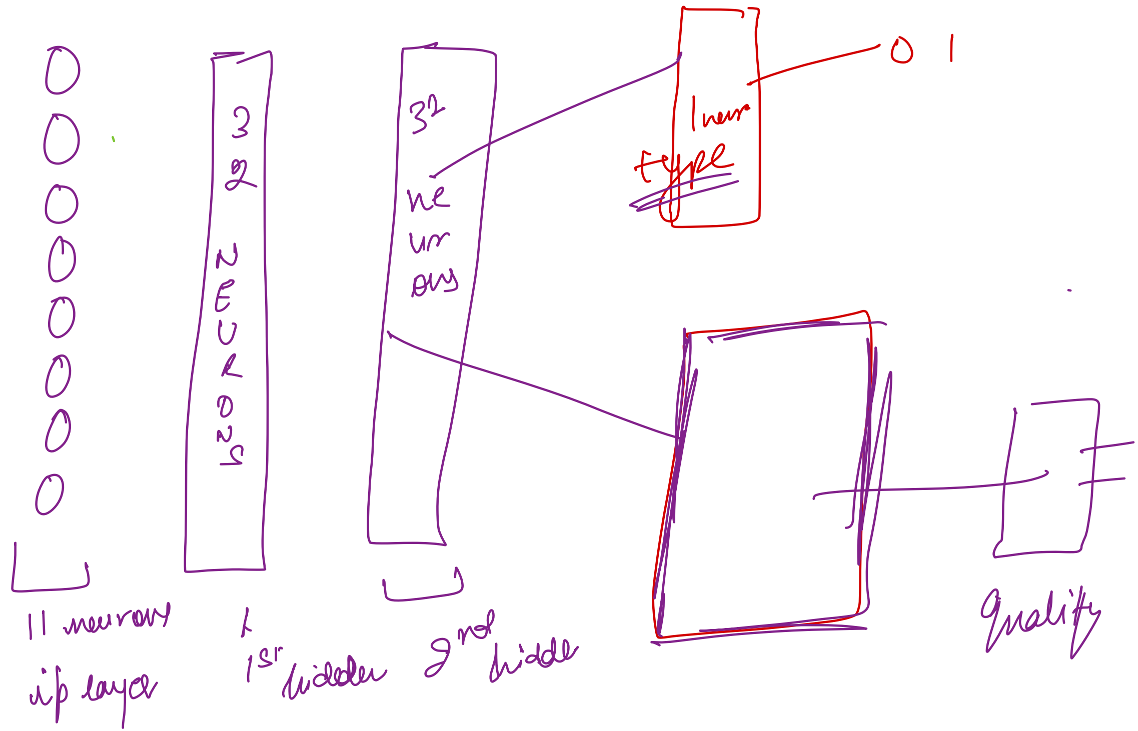


32 NEURONS
32 neurons
1 neur
type
0 1
11 neurons
i/p layer
1st hidden
2nd hidde
Quality

Callbacks → monitoring & improving the training process for ML.

Pre-built

1. Early stopping → Stop the training when there is no improvement
2. Model checkpoint → Save model or model weights at regular intervals
   - Save our best possible model so far.
   - resume your training from that checkpoint
3. CSV Logger → save metrics to a CSV file.
4. Reduce LR on Plateau → 0.1
5. Tensorboard → UI

Verbose = 1 → every detail of epoch

Verbose = 0 → no. detail.

Customise this behaviour

Custom Callbacks

→ On-epoch-begin

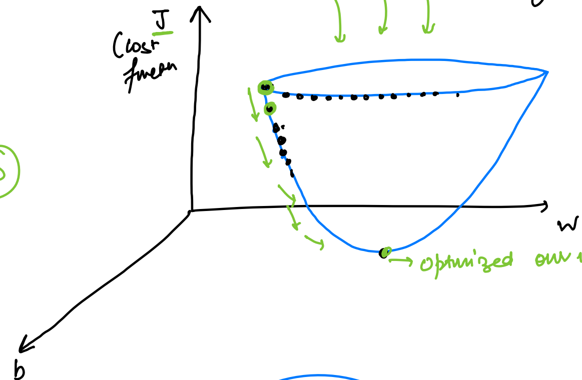
→ on-epoch-end

Optimizers

each [■] signifies what J

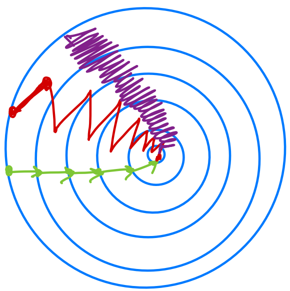
when all the data point (300) goes through the network

3D



→ optimized our weight & bias

2D



Batch GD

5 epoch to reach the minimal point

300 datapoints → whole

↳ 5x300 = 1500 calculation

Mini batch GD

100 datapoints Randomly , 100

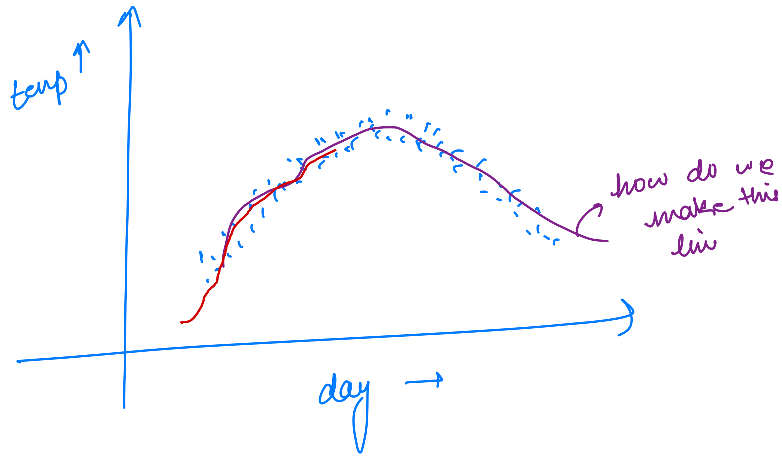
10 epoch to reach to the optimal point

100 x 10 epoch = 1000 calculation

How to reduce it's wavyness?

GD with momentum

$t(1) = 40$
$t(2) = 49$
$t(3) = 45$
⋮
$t(180) = 60$



weighted Avg (0th day)

$V(0) = 0$

Current value

$V1 = 0.9 \times V_0 + 0.1 \times t(1)$

$= 0.9 \times 0 + 0.1 \times 40$

$\Rightarrow 4$

0.9 & 0.1

$V2 = 0.9 \times (V1) + 0.1 \times (t_2)$

$= 0.9 \times 4 + 0.1 \times 45$

$V_t \rightarrow \beta \times V_{(t-1)} + (1-\beta) \times temp(t)$

dw

V_{dw}

$\beta = 0.9 \rightarrow 10$ days average

$\beta = 0.98 \rightarrow 50$ days average

$\frac{1}{1-\beta}$ days

0.5

$\frac{1}{1-0.5} = \frac{1}{0.5} = 2$ days

$W = W - \alpha \frac{\partial J}{\partial W}$ → updating of weights

$W = W - \alpha V_{dw}$

Noise

EWA

$V_{dw} = \beta V_{dw} + (1-\beta) dw$ → Acts as acceleration

Prev day's EWA of temp

Temp

friction

Velocity / momentum

momentum $= m v$